

Spectrophotometry and Beer's Law

Answer key with worked solutions

A. Understanding

1. Definitions.

(a) **Transmittance** $T = I \div I_0$, the fraction of light passing through the sample. **Absorbance** $A = -\log T = \log(I_0 \div I)$. (b) Absorption is **strongest** at $\lambda_{\max}$, so the method is most **sensitive**, and small errors in the wavelength setting change A least. (c) A **blank** contains the solvent and reagents but not the analyte; it is used to **zero** the instrument so that only the analyte's absorption is measured.

B. Absorbance and transmittance

2. Conversions.

(a) $A = -\log(0.500) = \mathbf{0.301}$ (b) $A = -\log(0.100) = \mathbf{1.000}$ (c) $T = 0.350$, $A = -\log(0.350) = \mathbf{0.456}$ (d) $T = 10^{(-0.200)} = 0.631$, so **63.1 %T** (e) $T = 10^{(-1.00)} = 0.10$, so **10 %T** (two decimal places in A give two significant figures in T)

C. Beer's law

3. Concentration.

$$c = \frac{A}{\epsilon b} = \frac{0.725}{(1.45 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1})(1.00 \text{ cm})} = \mathbf{5.00 \times 10^{-5} \text{ mol/L}}$$

4. Molar absorptivity.

$$\epsilon = \frac{A}{bc} = \frac{0.360}{(1.00 \text{ cm})(2.50 \times 10^{-5} \text{ mol/L})} = \mathbf{1.44 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1}}$$

5. Predictions.

A is proportional to both b and c ($A = \epsilon bc$). (a) Doubling b doubles A : **0.720**. (b) Halving c halves A : **0.180**.

6. Dilution.

$$\begin{aligned} \text{(a) } c &= \frac{0.370}{(1.20 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1})(1.00 \text{ cm})} = 3.08 \times 10^{-5} \text{ mol/L, so } \mathbf{3.08 \times 10^{-5} \text{ mol/L}} \\ \text{(b) Dilution factor} &= \frac{50.00 \text{ mL}}{10.00 \text{ mL}} = 5.000; c = 3.083 \times 10^{-5} \text{ mol/L} \times 5.000 = \mathbf{1.54 \times 10^{-4} \text{ mol/L}} \end{aligned}$$

D. Calibration and replicates

7. Triplicate unknown.

$$\begin{aligned} \text{(a) Mean: } \bar{A} &= \frac{0.334 + 0.338 + 0.331}{3} = 0.3343; \text{ sample standard deviation } s = 0.0035; \\ \text{RSD} &= \frac{0.0035}{0.3343} \times 100 \% = 1.05 \%. \text{ Report } \bar{A} = 0.3343 \pm 0.0035 (n = 3). \\ \text{(b) } c &= \frac{\bar{A} - 0.003}{0.0820 \text{ L/mg}} = \frac{0.3343 - 0.003}{0.0820 \text{ L/mg}} = \mathbf{4.04 \text{ mg/L}} \end{aligned}$$

8. Good practice.

(a) A single reading cannot show its own **precision**. Replicates give a **mean** (a better estimate) and a **standard deviation** (how closely the readings agree), and reveal any reading that is out of line. (b) The straight line is only known to hold **between** the standards; outside them, Beer's law may fail (especially at high A), so the result could be wrong. (c) A large intercept suggests a **systematic error**, for example an incorrect blank, a contaminated reagent or a dirty cell.

E. Challenge

9. Copper in brass.

$$\begin{aligned} c &= \frac{A}{\epsilon b} = \frac{0.409}{(395.0 \text{ L mol}^{-1} \text{ cm}^{-1})(1.00 \text{ cm})} = 1.035 \times 10^{-3} \text{ mol/L} \\ n(\text{Cu}) &= 1.035 \times 10^{-3} \text{ mol/L} \times 1.000 \text{ L} = 1.035 \times 10^{-3} \text{ mol} \\ m(\text{Cu}) &= 1.035 \times 10^{-3} \text{ mol} \times 63.55 \text{ g/mol} = 0.06577 \text{ g} \\ \% \text{ Cu} &= \frac{0.06577 \text{ g}}{0.1000 \text{ g}} \times 100 \% = \mathbf{65.8 \%} \end{aligned}$$